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Step 1 - Why Timers Exist in ARM Systems

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture and Cortex-A Generic Timer architecture documentation.

Learning Objectives

After this chapter you should understand:

- Why processors need timers
- Why software loops are unreliable
- Counter vs Timer
- Time Measurement vs Event Generation
- Why ARM introduced Generic Timer architecture

1. Why Does a Processor Need a Timer?

A processor executes instructions but has no inherent understanding of:

1 second
10 milliseconds
100 microseconds

The processor only knows:

Fetch
Decode
Execute

A timing mechanism is required to:

- Measure elapsed time
- Generate periodic events
- Detect timeouts
- Schedule activities

2. Why Delay Loops Are Not Reliable

Example:

```
for(i=0;i<1000000;i++);
```

Problems:

Frequency Dependency

1 GHz -> 1 ms
2 GHz -> 0.5 ms

Processor Dependency

Cortex-A53 ≠ Cortex-A72

Compiler Dependency

The loop may be optimized away.

Key Principle

Timing should come from hardware, not software loops.

3. Two Uses of Time

A. Time Measurement

Question:

How much time elapsed?

Example:

End Time - Start Time

Used for:

- Benchmarking
- Profiling
- Latency measurement

B. Event Generation

Question:

Has the target time been reached?

Example:

Generate interrupt after 10 ms

Used for:

- Periodic events
- Timeouts
- Scheduling

4. Counter vs Timer

Counter

Measures time.

Example:

100
101
102
103

Question answered:

What time is it?

Timer

Generates an event.

Example:

Current Count = 100
Target Count = 200

When:

Current Count >= Target Count

generate event.

Question answered:

Has the target time been reached?

Relationship

Counter
↓
Measures Time
Timer
↓
Uses Counter
↓
Generates Event

5. Deterministic Timing

Real-time systems require:

Predictability

not necessarily:

Speed

Good:

1 ms
1 ms
1 ms
1 ms

Bad:

0.5 ms
2 ms
0.8 ms
3 ms

6. Why ARM Introduced Generic Timer

Before Generic Timer:

Vendor A Timer
Vendor B Timer
Vendor C Timer

Different programming models.

ARM standardized:

System Counter
+
Architectural Timers
+
Architectural Registers

Benefits:

- Consistency
- Portability
- Multi-core support
- Virtualization support

Summary

- CPUs do not understand time.
- Software loops are unreliable.
- Hardware timers provide accurate timing.
- Counters measure time.
- Timers generate events.
- Deterministic timing is essential.
- ARM Generic Timer provides a standardized timing architecture.

Knowledge Check

1. Why can't a processor inherently measure milliseconds?
2. Why are delay loops unreliable?
3. What is the difference between a counter and a timer?
4. What are the two major uses of time?
5. Why did ARM introduce Generic Timer Architecture?